Shri. Khanderai Pratishthan's

DNYANSAGAR INSTITUTE OF MANAGEMENT & RESEARCH

Approved by AICTE, New Delhi, Affiliated to Savitribai Phule Pune University

PROJECT SYNOPSIS

On

CivicSaathi

Web-Based Civic Grievance Redressal and Community Portal

Name	Adarsh Pandey
Roll No	04

To

DEPARTMENT OF MASTER OF COMPUTER APPLICATION

Academic Year 2026-27

F.Y.MCA (SEM-I)

Guide name 1 - Prof. _____

Guide name 2 - Prof. _____

INDEX

Sr. No	Chapter	Page Number
1	Introduction	3
2	Existing System	4
3	Need for System	5
4	Proposed System	6
5	Objective of System	7
6	Scope	8
7	Module Specification	9
8	Operating Environment Hardware and Software	10
9	Technology	11
10	Bibliography	12

1. Introduction

Project Title: CivicSaathi: Web-Based Civic Grievance Redressal and Community Portal

Domain: Web Application Development / E-Governance / Information Systems

In towns and cities, people face small but serious daily problems. Roads have potholes, garbage bins overflow, water pipes leak, streetlights do not work, and drains get blocked. These problems make daily life hard for every citizen.

When such a problem happens, a citizen has to go to the municipal office, write an application on paper, or call a helpline that often does not answer. This is slow, and many complaints get lost.

CivicSaathi is a simple website with two separate logins: one for citizens and one for municipal officers. A citizen can register a complaint in about 60 seconds from a phone or computer, attach a photo of the problem, and track its status. Officers get a dashboard where they can see the complaint and photo, assign field teams, and mark the work as 'Resolved' with a note.

Key Features:

- **Two logins:** separate access for citizens and municipal officers.
- **Quick complaint form:** choose a category, write the address, and submit.
- **Photo proof:** take a live photo with the camera or pick one from the gallery.
- **Reference ID:** every complaint gets a unique tracking number.
- **Status tracking:** Pending Review, In Progress and Resolved.
- **Printable slip:** an A4 acknowledgement receipt with a QR code.

2. Existing System

At present, most municipal corporations accept complaints in a manual way. The citizen visits the ward office, fills a paper form, and hands it to a clerk. Some people also call a helpline number or send a letter.

How the present system works:

- The complaint is written in a paper register at the municipal office.
- A clerk reads it and passes it to the concerned department by hand.
- The citizen is not given any tracking number or update.
- To know the progress, the citizen has to visit the office again or call.

Problems in the existing system:

- **Manual paperwork:** paper complaints can be lost, damaged or forgotten.
- **No photo proof:** a written complaint does not show how big or urgent the problem is.
- **No tracking:** the citizen cannot see whether anyone is working on the issue.
- **Mixed complaints:** all types of complaints come to one place, so roads and electricity complaints get mixed up.
- **Wasted time:** citizens spend time and money visiting offices again and again.

3. Need for System

Because of the problems in the manual system, a simple and reliable online system is needed. Almost everyone now has a smartphone with a camera and internet, so reporting a problem online is easy and fast.

Why CivicSaathi is needed:

- **To stop loss of complaints:** every complaint is saved safely in a database.
- **To give clear proof:** a photo shows the real size and condition of the problem, so officers can decide what repair tools are needed.
- **To give transparency:** the citizen gets a reference ID and can see the status at any time.
- **To divide work by department:** complaints go to the right department (Roads, Water, Sanitation, Electricity) so officers see only their own work.
- **To save time:** a citizen can complain in about 60 seconds without leaving home.
- **To create accountability:** an officer must write a remark about the work done before closing a complaint, so a record is always kept.

4. Proposed System

CivicSaathi is a dual-portal web platform. One portal is for citizens and the other is for municipal officers. It is simple, practical and works on both mobile and desktop browsers.

How the proposed system works:

- The citizen signs up or logs in.
- The citizen fills the complaint form and attaches a photo.
- The complaint is submitted and a unique Reference ID is created (for example REP-2026-X8K9).
- The municipal officer opens the Department Console and sees the complaint with its photo and address.
- The officer changes the status to 'In Progress' and later to 'Resolved', writing a remark.
- The citizen sees the updated status and officer remarks, and can print the receipt slip.

Benefits of the proposed system:

- Faster complaint submission and quicker action.
- Photo evidence with every complaint.
- Full status tracking and a clear history record.
- Complaints are sorted by department automatically.

5. Objective of System

Objectives of the project:

- To create a secure web platform that keeps Citizen access and Municipal Officer access separate.
- To give a quick 60-second complaint form with category selection and address input.
- To let citizens attach a photo using the live camera or the phone gallery.
- To create a unique Reference ID for every complaint so it is easy to track.
- To give officers a dashboard to filter complaints by department and update their status.
- To create a printable acknowledgement slip with a QR verification code.

Expected Outcomes:

- A working web portal with two separate logins.
- Correct complaint reporting with real photos and landmark addresses.
- Fast and clear tracking from submission to resolution.
- Official printable receipts with QR codes for records.

Conclusion:

CivicSaathi is a practical and reliable solution to an everyday public problem. It does not make unrealistic promises like complex artificial intelligence. It stays focused on what municipal offices really need: speed, accountability, photo proof and a clear digital record.

6. Scope

In Scope (what the project does):

- Citizen registration, login and profile management.
- Municipal officer login with department selection.
- Complaints in 5 categories: Road Potholes, Garbage Dumps, Water Leaks, Streetlights and Drainage.
- Photo upload using the camera or the gallery.
- Live status badges: Pending Review, In Progress and Resolved.
- Department-wise filtering for municipal officers.
- Safe cancellation or deletion of pending complaints by the citizen.
- Printing an official A4 acknowledgement receipt.

Out of Scope (what the project does not do):

- It is **not** an emergency system. It does not replace 112, Police or Fire Brigade services.
- It does **not** use costly or error-prone AI or machine learning. Photos uploaded by citizens are checked by municipal staff.
- It does **not** handle court cases or private property boundary disputes.

7. Module Specification

The system has five core modules, explained in the table below.

Module	Purpose	Main Functions
1. User and Role Authentication	Keeps data safe and separates citizen and officer rights.	Citizen signup/login with email, password and phone; officer login with department and Employee ID; protected pages so citizens cannot change municipal records.
2. Grievance Registration with Photo	Lets citizens report a problem with clear proof.	Category picker; live camera photo; gallery picker with preview; address and landmark input.
3. Citizen Tracking and History Dashboard	Gives the citizen full view of their complaints.	List of all complaints; colour badges (Pending = Yellow, In Progress = Blue, Resolved = Green); officer remarks; cancel button for pending or duplicate complaints.
4. Municipal Department Action Console	Work desk for engineers and ward officers.	Department filter; view photo and address; one-click status change; officer must write remarks before marking Resolved.
5. Digital Receipt and QR Slip	Gives the citizen official proof of submission.	Automatic unique Reference Code; Print Slip button for A4 receipt or PDF; QR code with verification link.

8. Operating Environment Hardware and Software

A. Hardware Requirements

Part	Requirement
Client (citizen / officer)	Any smartphone, tablet, laptop or desktop with at least 2 GB RAM and internet connection.
Server	Any computer or cloud host with 2 CPU cores, 2 GB RAM and 10 GB disk space.

B. Software Requirements

Part	Requirement
Operating System	Windows 10/11, Linux, macOS, Android or iOS.
Web Browser	Google Chrome, Microsoft Edge, Mozilla Firefox or Apple Safari.
Development Tool	Visual Studio Code (VS Code).
Runtime and Frameworks	Node.js (v18 or above), React 19, TypeScript, Express, Tailwind CSS.
Database	MySQL 8.0 or above (relational SQL database).

9. Technology

The project follows a standard **3-Tier Web Architecture**.

Tier 1: Presentation Tier (Frontend)

- Built with HTML, CSS and JavaScript pages served by Flask templates.
- Works well on mobile browsers and desktop computers.

Tier 2: Application Tier (Backend)

- Built with Python and the Flask framework.
- Handles login, complaint submission, status updates and report queries.

Tier 3: Database Tier

A MySQL relational database (connected to Flask using Python) with 3 tables:

Table	What it stores
users	User id, name, email, password, role (citizen or officer) and department.
complaints	Complaint id, Reference ID, user id, category, description, address, photo, status, officer remarks and time.
complaint_history	Status change log: officer id, old status, new status and remarks.

Data flow: Citizen logs in → fills form and attaches photo → submits → Reference ID created → officer sees it in Department Console → sets status to Resolved with remarks → citizen sees update and prints slip.

10. Bibliography

The following websites and documents were used for reference:

1. React Documentation – https://react.dev
2. TypeScript Documentation – https://www.typescriptlang.org/docs
3. Node.js Documentation – https://nodejs.org/docs
4. Express.js Guide – https://expressjs.com
5. Tailwind CSS Documentation – https://tailwindcss.com/docs
6. MySQL 8.0 Reference Manual – https://dev.mysql.com/doc
7. MDN Web Docs (Camera access and file upload) – https://developer.mozilla.org
8. W3Schools Tutorials (HTML, CSS, SQL) – https://www.w3schools.com